

Student Name: Cuong Dang

Course: ITAI-1378-Comp Vision-Artificial Intel

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L05 - Chihuahua or Muffin with CNN

1. CNN Architecture

In our last workshop, we used a traditional neural network called an MLP. To read an image, an MLP must flatten the 2D pixel grid into a single long 1D line of numbers. This step is a major flaw because it destroys all spatial relationships between pixels. The network becomes blind to where things are in space. In contrast, our new Convolutional Neural Network (CNN) keeps the image in its original shape. It passes small 3D filters over the color channels to learn patterns step-by-step. The first layers find basic edges and lines. The middle layers find textures and small shapes. The deep layers combine everything to recognize complex objects like a dog's snout or a muffin's round shape. Because the same filters slide across the whole image, CNNs have fewer parameters and can find objects no matter where they shift in the frame.

2. Model Performance

Our notebook resized the images to a uniform 64x64 grid and grouped them into batches of 32. We trained the network for 10 epochs using Cross-Entropy Loss and the adaptive Adam optimizer at a learning rate of 0.001. While the model achieved a high validation accuracy, the final visual grid showed that the network still makes systematic mistakes. Specifically, the model often fails on muffins that contain dark clusters of blueberries. To a localized convolutional filter, these dark spots create a mathematical signature that looks exactly like the eyes and nose of a Chihuahua. Similarly, when a Chihuahua curls up tightly inside a tan blanket, its shape, color, and texture distributions perfectly mimic a pastry wrapper. These errors happen because the CNN does not have human semantic understanding; it only looks at statistical pixel correlations.

3. Comparison

The CNN performs much better than our old MLP model in terms of accuracy and training time. The MLP hit an early performance wall because flattening the pixels made it ignore spatial shapes. The CNN easily extracted clean geometric features and avoided early learning flatlines. In terms of training time, 3D convolutions require a lot of matrix calculations, but they are highly parallel. Running the CNN on a hosted Nvidia T4 GPU in Google Colab lets these highly parallel operations finish very fast, significantly accelerating

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our training loop compared to a standard CPU. Since the CNN uses weight sharing, its overall file size and parameter count are much smaller than a fully connected MLP, saving computer memory while delivering a much higher score.

4. Challenges and Solutions

We faced two major technical challenges during this lab. First, the default dataset code sets the data loader to use 4 worker processes. This triggered a warning because Google Colab only provides 2 CPU cores on its free tier. Forcing 4 workers caused a resource bottleneck that threatened to freeze the runtime. We solved this by changing the code to 2 workers, which optimized our data pipeline. Second, the visual evaluation script was hardcoded to a fixed 5x6 grid but tried to loop through the entire validation folder. If the folder had more than 30 images, this would cause an automatic crash. We patched the loop logic to use a dynamic maximum bound of 30, ensuring smooth execution. Additionally, to meet the experiment requirements, we successfully tested a lower learning rate, swapped Adam for an SGD optimizer, and turned off Dropout to analyze how the architecture reacts to overfitting.

5. Real-World Applications

The mathematical concepts of CNN feature extraction are highly valuable in commercial industries. For example, in a massive e-commerce company that processes millions of new product photos daily, passing every image straight into a heavy AI model is too expensive and slow. A practical engineering solution is to use fast, traditional scripts to crop, clean, and filter out bad photos first. Once the bulk data is organized, the clean images are passed to the CNN for final classification, balancing speed with accuracy. Additionally, these models are widely used in medical imaging to help doctors automatically separate benign tissue from malignant tumors in high-resolution X-ray or MRI scans.

6. Ethical Considerations

Developing and deploying computer vision models requires strict ethical standards. First, a model is only as good as its data. If an image dataset is scraped carelessly from narrow web sources and lacks diverse lighting or environments, the model will fail or show systemic biases when used in the real world. Second, we must be careful with automation bias. Humans have a psychological tendency to trust machine output blindly. If a CNN outputs a wrong prediction due to a minor change in lighting, and humans do not double-

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check it, critical failures can go unnoticed. Maintaining human oversight is an absolute ethical necessity.

References

1. Elgendy, M. (2020). *Deep Learning for Vision Systems*. Manning Publications. (Chapter 3: Image Classification with MLPs and CNNs).
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. (Chapter 9: Convolutional Networks).